Supplementary material for PLECTA: Overlap-aware instance reconstruction of dense filament networks from binary masks

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S1 Reconstruction: cost function, gates and schedule

This section gives the arithmetic behind Section 2.4 of the article. Values are those of the released configuration; nothing here was tuned on held-out or locked data.

S1.1 Skeleton graph

The mask M is binarised at > 0, skeletonised, and spurs no longer than 3 px are removed iteratively. Skeleton pixels of degree at least three form junction clusters, and adjacent clusters merge into nodes: connectors up to 5 px are absorbed into one node, connectors up to 14 px merge nearby nodes while remaining real arms, and free debris up to 2 px is absorbed. Each remaining connected path is stored as an ordered arm carrying one stub at each end.

S1.2 Stub frames

A stub frame holds its tip, outward tangent and signed curvature, estimated by a least-squares arclength polynomial fit read outward from the tip. The fit is quadratic when the span reaches 18 px and linear otherwise, and a frame counts as reliable only with at least three samples spanning at least 3 px. Local frames use 24 px of support; once links form chains that support expands to 55 px, so a stub too short to have a direction of its own borrows one from the chain it has joined.

S1.3 Weights and scales of the pairing cost

The cost function itself is stated in the article, which also gives the reasoning behind its three deliberate properties. What follows are the values it is evaluated with.

Junction and gap links use separate weight sets, because a junction link is scored on direction alone while a gap link must also answer for the distance it spans. The curvature term carries a constant of 30, which places curvature, in px^{-1}, on the scale of the angular terms; without it a term in reciprocal pixels and a term in radians could not be added. Junction links set $w_\ell = 0$, a junction having no gap to pay for, and ℓ_0 and d_0 set the length scale of the gap penalty and the width of the chord fade respectively.

S1.4 Prices, gates and admissibility

Leaving a stub unmatched is priced rather than forbidden, and the price p_i also fixes admissibility: an edge is offered only when $C_{ij} < p_i + p_j$. Exact maximum-weight partial matching then maximises $\sum(p_i + p_j - C_{ij})$ over disjoint pairs, which minimises total link cost plus p_i for every stub left free.

Table S1. Cost weights and scales, read from the frozen configuration that produced every published number. Junction and gap links are scored with separate sets; the gap set alone carries a length term, a junction having no gap to pay for.

Symbol	Term	Junction	Gap
w_d	reversal angle θ	0.6	0.2
w_t	chord turns $\varphi_a + \varphi_b$	1.2	1
w_κ	curvature continuity	1	0.5
w_ℓ	length d/ℓ_0	—	0.15
ℓ_0	length scale (px)	—	60
d_0	chord-fade scale (px)	4	3
p_i	price of leaving a stub free	0.62	0.3

The admissibility limit is not a separate threshold. An edge with $C_{ij} \geq p_i + p_j$ contributes nothing positive to the objective, so a maximum-weight *partial* matching would never select it, and declining to offer it removes an edge that could not have been chosen.

At the configured prices junction stubs carry $p_i = 0.62$, so a junction edge needs $C < 1.24$. Gap stubs carry $p_i = 0.30$, so a gap edge needs $C < 0.60$, and must in addition satisfy $d \leq 85\,\text{px}$, $\theta \leq 0.40\,\text{rad}$ and $\varphi \leq 0.28\,\text{rad}$ at each end. The gap gates are the tighter set because a gap link asserts a continuation across evidence that is missing rather than merely ambiguous. The values were fixed on development scenes.

S1.5 The round schedule

Rounds $r = 0, \ldots, 8 - 1$ apply a schedule fraction $\alpha_r = 0.85 + (1 - 0.85)\,r/7$ that scales the unmatched prices and the two gap angle limits, and nothing else. Scaling a price *down* admits fewer edges, admissibility being $C_{ij} < p_i + p_j$, and scaling an angle limit down narrows the gap gates, so early rounds are the conservative ones and are deliberately under-linked: while the frames are least reliable, only pairings that survive the reduced prices and narrowed gates are committed, and the constraints relax towards their configured values as the chains assembled so far sharpen the frames. The final round, $\alpha_7 = 1$, is the only one run at the configured parameters, and is the one returned.

In full:

1. Skeletonise M, prune spurs, merge junction clusters into nodes, and store the remaining paths as arms with a stub at each end.
2. For $r = 0, \ldots, 7$, re-solving both matchings from scratch and carrying only the chain structure forward from round $r - 1$:
   (a) Re-fit every stub frame along the chain it now belongs to.
   (b) At each node, form candidate edges between stubs of different arms with finite frames and cost, keep those admissible at α_r, and solve one maximum-weight matching per node.
   (c) From the still-unmatched reliable stubs, form gap candidates on different arms and different nodes that pass the distance gate and the α_r-scaled angle gates, and solve one global maximum-weight matching over the admissible ones.
3. Return the links of round $r = 7$.

S1.6 Open-curve guard

Instances are assumed to be open projected curves, and a guard enforces it: after each of the two matchings, any matched component whose links close a ring has its longest link removed. On the 84 development scenes the guard never fired, with zero links removed, so it protects against a configuration the data did not produce rather than shaping the result. Closed or looping filaments are outside the scope of this paper.

S1.7 Why the junction is solved as a whole

Joint and sequential solutions coincide at plain crossings and diverge only as nodes grow dense. Over 20 development scenes they never disagreed at nodes of degree 2–4 ($n = 542$), disagreed at 22 % of nodes with 5–8 stubs ($n = 187$) and at 69 % of nodes with 9 or more ($n = 233$), where the joint solution was better by a median of 0.90 in total cost.

That comparison holds one node's costs fixed. Run inside the rounds, where each matching sets the frames the next is scored against, a cheap early choice propagates and the difference is no longer confined to dense nodes.

S1.8 Instance output

Connected components of accepted arm links define instances. PLECTA paints every arm pixel and every junction cluster touched by a chain into that instance, so intersecting instances share crossing pixels. An isolated one-arm component shorter than 6 px is omitted. Accepted gap links determine identity but are not painted into the fixed centreline output. All effective parameters, including values previously inherited as code defaults, are materialised in the released JSON configuration.

S2 Depth stage: evidence and solver

This section gives the detail behind Section 2.6 of the article.

S2.1 Sampling geometry at a crossing

Where two instance centrelines intersect, a short arc of each is taken through the crossing (the shared *core*, a flat 7.5 px of arc, the same at every crossing), together with the *flanks* on either side of it. Flank samples are kept clear of the other filament's stroke, because a sample taken inside it measures the union of two rods rather than one of them. A crossing with too few usable flank samples abstains rather than being decided on thin evidence.

The core length was previously sized from the two filaments' radii so that it spanned both silhouettes. It is now the constant above, and no per-rod radius enters the evidence path at all, so a radius estimate can no longer move a depth decision. Radii re-enter only downstream, in the non-interpenetration constraints that set metric height.

S2.2 One channel, and its noise floor

One quantity is read on core and flanks alike: median intensity. Write m_i and m_j for the two filaments' flank medians and m_c for the pooled core median. The evidence at the crossing is the score

$$s \;=\; 2\,\frac{|m_c - m_j| - |m_c - m_i|}{\max\bigl(|m_i - m_j|,\; 2\hat{\sigma}\bigr)}, \tag{S1}$$

whose sign names the upper filament – the one whose flanks the core resembles more closely – and whose magnitude $|s|$ is the weight that decision carries. A crossing abstains when $|s| < 0.40$; the threshold is set directly in score units rather than as a band around a probability. Here $\hat{\sigma}$ is a noise estimate belonging to that crossing alone: 1.4826 times the median absolute deviation of the within-rod flank residuals, each filament's flank samples being centred on their own median and the two residual sets pooled. It therefore measures the scatter within a rod's own appearance, not the difference between the two rods.

A probability p is still written into the per-crossing record so that the output keeps its schema, but it is a monotone relabelling of s. It is neither a calibrated confidence nor the quantity anything is decided or weighted on, and should not be read as one.

Why the denominator is a noise floor

By the reverse triangle inequality the numerator of (S1) is bounded in magnitude by $|m_i - m_j|$, so a denominator equal to that separation drives the ratio to ± 1 whenever the two filaments differ at all. The previous form of this rule floored the denominator at a fixed 0.02 grey levels, far below the noise of any of these images. The consequence was structural rather than statistical: $|s| \geq 1.98$ then exceeded any abstention threshold, so the rule could not abstain on such a crossing whatever the evidence actually was. Measured, 32.3 % of synthetic and 61.5 % of real crossings were saturated this way, and among the saturated ones those whose separation fell below the noise scored about 0.70 accuracy against about 0.93 for those above it. Flooring at $2\hat{\sigma}$ instead makes a confident score mean *separated by more than the noise* rather than merely separated, and restores the stage's ability to abstain where it should.

The gradient channel, and why it was removed

An earlier version of this stage read a second quantity beside intensity: median gradient magnitude sampled on each filament's own two edges at centreline $\pm r\,\mathbf{n}$, the axis gradient being near zero and therefore uninformative. The argument was that the upper filament's edges stay crisp through the crossing while the lower filament's are overwritten. It did not survive measurement. The channel decided 0.2 % of real crossings, agreed with the intensity channel at chance ($\kappa \approx +0.09$), and scored 0.54 to 0.59 standing alone, which is chance. The reason is visible in its own sampling geometry: within the core those $\pm r$ offsets fall inside the *other* filament's stroke, so the quantity is contaminated exactly where it would have had to be read. The channel is removed, and with it the gradient image it required, the two channel weights that combined it with intensity, and the logistic – with one channel there is nothing to combine, and s is already the decision variable.

What the replacement is worth

The two rules were compared on a held-out set of 20 optically rendered synthetic scenes carrying 1342 crossings, whose seeds are disjoint from every set used to develop either rule and to choose between them. Its constants are a weaker claim: the flat core half-size (7.5 px) was settled by a sweep run on this same set, and the abstention threshold's flatness was checked on it, so with respect to those two values the set is not held out. Neither was invented here – 7.5 px is exactly what the previous rule's radius-free fallback already computed, and both values sit on a plateau rather than a peak, a flat core being no worse than oracle per-rod radii anywhere over 6–14 px and the accuracy unchanged across a band of thresholds around 0.40 – so nothing below turns on either value, but the comparison is in-sample with respect to them.

At matched coverage, meaning both rules made to answer as often as the shipped one does at 83.0 %, the one-channel rule matches the two-channel rule's accuracy: the difference is +0.0045, 95 % CI $[-0.0186, +0.0235]$, which is no difference, and no claim of higher accuracy is made. One reading does go against the replacement and is reported rather than omitted: forced to decide *every* crossing, including those it would decline, the one-channel rule is worse by -0.0142, CI $[-0.0313, +0.0007]$. That is the expected direction, because abstention is exactly where an ordered confidence pays, and removing it removes the advantage; it is also not how either rule is run. What improves is the ordering of the stage's own confidence. The area under the curve of correctness against $|s|$ rises from 0.6531 to 0.7586, a gain of +0.1056, CI $[+0.0659, +0.1425]$. That matters because $|s|$ is the edge weight the global ordering below is solved with: better-ordered confidence means contradictions are broken at the genuinely weakest relations rather than at arbitrary ones. It also converts into accuracy wherever more abstention is acceptable, by +0.0054 at 82.5 % coverage and +0.0121 at 80 %. The claim is therefore the same accuracy with better-ordered confidence, from a simpler evidence path.

A second channel, axial continuity along each filament, was built and measured on the same held-out set. It was inert – -0.0008, CI $[-0.0059, +0.0048]$ – and is deliberately not included.

S2.3 One order for the whole field

Each decided crossing contributes a directed edge weighted by $|s|$ itself, oriented toward the relation the sign of s names. The score is the primitive: the reported p is a relabelling of it and enters no computation here. Within each connected component of the crossing graph one total order over its instances is derived so as to maximise the total satisfied weight, and every relation in the component is then re-read from that order: the ones the order disagrees with are flipped rather than discarded, and both the flip and the original local score are recorded, so a reader can see how much the global correction moved. Which relations that turns out to be is the subject of the paragraph below.

Maximising satisfied weight is the minimum-weight feedback arc set objective, but only components of at most 14 instances attain it, by dynamic programming over subsets; larger ones get a greedy order refined with adjacent swaps, which carries no optimality guarantee. Because the order is re-derived for the component as a whole rather than only where relations conflict, a greedily ordered component can return a relation reversed that nothing contradicts, which an exact solution never does – a reversal lying on no cycle only loses weight. Fed ground-truth crossings with their true directions, input that is acyclic by construction and that an exact solver would leave untouched, the pass still reverses 24 relations across the 20 held-out scenes, capping layer agreement at 0.886 where it should reach 1.000; a condense-then-order variant, which orders the condensation of the strongly connected components and reorders only inside genuine cycles, reverses none of them but measured worse on the actual noisy predictions and is not what ships.

S2.4 Layers and metric height

The corrected relations form a directed acyclic graph whose longest path fixes the smallest number of layers consistent with them; filaments that never conflict share a layer, so the layer count follows from the ordering rather than from the number of instances. Each instance is then placed at a height respecting its order that keeps filaments from interpenetrating at their measured radii, with the stack compacted so the film is as thin as those constraints allow.

S2.5 Configuration

Every tunable quantity of both stages (the cost weights and gates above, the core and window sizes, the abstention threshold in score units, and the exact-solver size limit) is carried in a single released parameter file, grouped by the stage it belongs to, with the 2-D grouping section checked against the frozen record that produced the published numbers. Table S1 gives that file's 2-D half; Table S2 gives its depth half, so both are shown rather than asserted. Every value in either table, and every parameter this section quotes in passing, is read from the released file when this document is built.

S3 Measures

Every measure this paper reports is defined here, with what it rewards and its blind spot; none reads the rendered width of an instance.

Table S2. The depth stage's own entries in the released parameter file. The abstention threshold is in score units, not a band around a probability, and the core is flat: no per-rod radius enters the evidence path.

Quantity	Value	What it sets
core (px)	7.5	half-length of the shared crossing arc; flat at every crossing, so no per-rod radius enters the evidence path
window (px)	22	arclength sampled either side of a crossing for evidence
min. flank	6	fewer usable flank samples and the crossing abstains
abstain $\|s\|$	0.4	abstention threshold, in score units, not a probability band
exact limit	14	components to this many nodes are ordered exactly, larger ones greedily

The common-fragment set, shared by the first five measures below. The mask is skeletonised, spurs of at most 3 px are pruned, skeleton pixels with three or more 8-neighbours are deleted, the remaining 8-connected pieces are labelled, and pieces shorter than 6 px are discarded, so one fragment is one unambiguous arm between crossings. The partition reads neither the reference nor any prediction, so every method compared is cut into the same fragments. Each then takes a reference and a predicted identity by one rule, applied to both sides alike: the fragment is assigned to the instance whose mask overlaps the largest number of its pixels, the vote counted against each instance's full mask rather than a flattened label image, so an instance that overlaps another keeps its pixels, provided that the winning overlap covers at least 30 % of the fragment; failing that, it falls back to the most frequent nearest instance label within 6 px of the fragment, read from a deterministically flattened field; failing both, it stays unassigned and is scored as its own singleton cluster, on the reference side and the prediction side equally, so every fragment enters the scored partition. Ties at any step resolve to the lowest instance id, and the flattening order is fixed, so the assignment is deterministic. Because junction pixels are deleted first, no fragment contains one: none of these measures can see whether an emitted crossing pixel is given one owner or two, only whether the arms meeting there are grouped as the reference groups them.

Sensitivity to the scorer's constants. Because the four constants above are shared with the method (the article's metrics section), the stored predictions of all four methods were re-scored under one-at-a-time perturbations of each constant, with the grid frozen in writing before any result was computed: spur pruning 2–4 px, minimum fragment length 4–8 px, overlap threshold 0.20–0.50, fallback distance 3–9 px. Nothing was re-run and no configuration was changed in response. Table S3 reports PLECTA's mean F_1 and its plain paired margins over each comparator at every perturbation, over the 50 degraded comparison scenes. Across the 10 combinations PLECTA's mean F_1 spans 0.713–0.859, and the smallest paired margin over any comparator under any perturbation is 0.115. The level moves only with the minimum fragment length, and in the same direction for every method, because admitting shorter fragments adds hard, low-evidence units for everyone; the other three constants move the mean by less than 0.002, and no perturbation changes the ordering of the four methods.

Common-fragment pairwise $\mathbf{F_1}$, the primary endpoint. A pair of fragments is a true positive when it shares both identities, a false positive when the prediction joins fragments of different reference filaments, a false negative when it separates fragments of one:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad F_1 = \frac{2PR}{P+R}. \tag{S2}$$

A clustering measure of the Rand family [1–3], permutation-invariant in the instance labels; it rewards holding a filament together across every crossing and gap it passes. Blind spots: it is

Table S3. Scorer-constant sensitivity. Stored predictions re-scored under one-at-a-time perturbations of the four common-fragment constants; Δ columns are plain paired mean margins, PLECTA minus comparator, over the scenes both methods completed. Sensitivity reporting only, no inference, and the published constants were not changed in response.

Scorer constants	PLECTA F_1	Δ greedy	Δ DNAi	Δ GraFT
published constants	0.822	0.145	0.478	0.339
spur prune 2 px	0.822	0.145	0.478	0.339
spur prune 4 px	0.822	0.145	0.478	0.339
min fragment 4 px	0.713	0.115	0.388	0.267
min fragment 8 px	0.859	0.149	0.507	0.358
overlap threshold 0.20	0.822	0.145	0.478	0.339
overlap threshold 0.40	0.822	0.145	0.478	0.339
overlap threshold 0.50	0.822	0.145	0.478	0.339
fallback distance 3 px	0.824	0.145	0.478	0.340
fallback distance 9 px	0.821	0.145	0.478	0.340

quadratic in fragmentation, so a filament arriving in n pieces carries $\binom{n}{2}$ pairs and a broken mask is charged many times for one missed join, and it weights long, much-crossed filaments far above short ones.

Adjusted Rand index [4] counts the same pairs corrected for chance, which matters because with a hundred filaments per scene almost every pair genuinely belongs to different filaments and the uncorrected index sits near 1 for anything. Blind spot: redundancy, since over the comparison set it never departs from pairwise F_1 by more than 0.019, so it confirms rather than adds.

Variation of information [5], in bits, lower better, is the conditional-entropy distance between the partitions: $\mathrm{VI}_{\mathrm{split}} = H(\text{prediction} \mid \text{reference})$ is paid for cutting one filament into several, $\mathrm{VI}_{\mathrm{merge}} = H(\text{reference} \mid \text{prediction})$ for fusing several into one, and reporting both is what separates a method that divides from one that fuses. Blind spot: each component alone is degenerate, since a method merging everything never splits anything and scores a perfect zero on the split component, so neither is ever quoted singly.

Detection $\mathbf{F_1}$, the measure DNAi reports, matches predicted to reference *objects* greedily by highest intersection over union, stopping below 0.1, DNAi's own shipped threshold, with precision over predicted and recall over reference objects. It uses a symmetric 2 px placement tolerance, under which a reference pixel counts as found when the prediction passes within it and vice versa, the intersection being the smaller of the two counts. The tolerance is necessary: two independently read one-pixel centrelines two pixels apart intersect in *nothing*, so a strict overlap reports placement rather than shape, a problem the thin-structure benchmark FISBe states as a design rationale [6]. Counting objects makes it the most forgiving reading of a fragmented mask and the least discriminative measure here. Blind spot: an arm swap, which it scores 1.000 where the pairwise score gives 0.000.

Crossing Fidelity states the contribution directly. At every junction node the reference induces a partition of the incident arms, two together exactly when the reference gives them the same filament, and so does the prediction; Crossing Fidelity is the fraction of crossings at which the two partitions are identical. Nodes are the junction clusters of the same pruned skeleton and a node's arms are the fragments 8-adjacent to it. Only nodes whose arms carry at least two distinct reference filaments are scored, since a node where every arm belongs to one filament is a skeletonisation artefact and scoring it would reward doing nothing. It is an in-house diagnostic, not an established metric. Blind spots, both of which govern its use: it is far less discriminative than the pairwise score, so it says what was solved and is never a ranking axis; and crossings carry no information about *gaps*, so it is silent on the cost the pairwise score pays for bridging them. Sharing its skeleton and assignment with the pairwise score makes it a sharper view of the same evidence rather than independent confirmation, and it is always printed beside its all-unpaired baseline.

Join $\mathbf{F_1}$ counts decisions rather than pairs, for the real fields where the two mask sources fragment differently. A *decision* is a pair of distinct fragments whose endpoints come within 85 px, the gap this method will bridge; reference and prediction each say join or not, and precision, recall and their harmonic mean are taken over decisions. Being linear rather than quadratic in fragmentation, and conditioned on the mask, it asks two mask sources comparably hard questions, and splitting and merging appear separately as low recall and low precision. Blind spot: a fragment pair the mask never brought within reach counts against nobody, so it cannot see a filament the mask lost entirely, and it is always quoted with the number of decisions each mask posed.

Reference-instance recovery is the fraction of reference filaments recovered whole by one predicted instance. The dominant predicted instance of a filament owns most of its fragments; coverage is that count over the filament's fragment count, purity the same count over every fragment that instance owns anywhere, and a filament counts as recovered when the dominance is *mutual* and both reach 0.8. Purity is what stops a method that merges everything from scoring 1.0, and the measure weights every filament equally, complementing the pairwise score's quadratic weighting. Blind spot: the two thresholds, which make it insensitive to how badly a failing filament was reconstructed.

Completeness, correctness and quality characterise the input *mask*: no reconstruction is opened to compute them and no method can change them. Reference axis and test mask are both skeletonised and each tested against the other dilated by the tolerance. Completeness is tolerant recall of the reference axis, correctness tolerant precision of the test axis, and quality the tolerant Jaccard index $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$ with TP the smaller of the two matched counts, so quality cannot exceed either component. Quality is also GraFT's published Jaccard index in its intended reading. Blind spot: the tolerance carries the measure, since at zero tolerance it reports raster placement rather than geometry, so it is never quoted without one.

Shared-pixel precision and recall are precision and recall over the pixels the reference gives to more than one instance, computed on the raw instance masks. They exist because the pairwise measure discards junction pixels and cannot see whether a crossing pixel was marked as jointly owned at all: only a method emitting overlapping layers can score above zero on the recall, and one emitting a hard partition scores exactly zero by construction. Blind spot: neither component distinguishes a pixel shared for the right reason from one shared because a chain was painted with a whole junction cluster, which is the over-marking the article's crossing-representation results report.

The runtime scaling exponent is the slope of an ordinary least-squares fit of log seconds on log foreground pixels, so an exponent b means cost grows as the b-th power of the amount of ink. It is reported instead of absolute time because these are independently optimised codebases, where absolute time measures implementation as much as algorithm. Blind spot: it describes a median trend and is silent on the tail, so it is quoted beside the per-density medians and the censored-scene count; censoring biases it downwards for any method whose slowest scenes did not finish, which is why GraFT's is a lower bound.

Two measures were computed and are not endpoints. GraFT's *filament matched coverage* asks whether each reference filament was matched one-to-one to some detection, and PLECTA ranks *below* both comparators on it: at the densest stratum of GraFT's own generator, 0.798 against 0.998 and 0.900. It is not an endpoint because it counts existence rather than correctness: a method emitting more objects than the scene contains has a spare for every filament, each method's matched coverage equals the smaller of one and its emission ratio to within 0.16, and requiring a detection to cover half the axis it claims restores the order. *Centreline Dice* [7] was tried as an alternative matching criterion and not adopted: it tests each side's skeleton against the other's *volume*, and a centreline grouping has none, so it degenerates into the placement test the tolerance exists to correct. No clDice result is reported in this paper.

S4 Comparator configuration

The article's comparator section states each adaptation and its cost; this is the mechanism behind it, for a reader checking that the comparison was run fairly. **DNAi** clusters junction pixels by single linkage and erases a disk at each cluster. At its released 10 px linkage distance, crossings closer than that chain into one cluster, which at 60 % coverage reaches 89 px across and is replaced by a single small erasure disk, leaving a whole tangle as one instance, which is why its shipped default is not a fair measure of it. Reducing the distance to 2 px on development scenes raised development F_1 and cut variation-of-information merge by about a factor of three, confirming the mechanism; the tuned configuration is the one reported. Two structural choices account for the residual difference: it commits to a pairing at every junction of degree two to four with no cost threshold, so it cannot decline a continuation the way a priced unmatched option can, and it attempts no junction above degree four, resolving 0.016 of the crossings above it. Those are about a tenth of the crossings, but they carry 37.2 % of same-filament fragment pairs on the degraded masks, one such junction severing every pair that spans it. **GraFT**'s errors are divisions rather than merges (variation-of-information split 1.166 against merge 0.392), which follows from a greedy longest-first path cover that cannot revise an early commitment once the scene collapses into one connected graph, as it does above 30 % coverage.

S5 Depth measures

The measures of Section S3 score which centreline fragments belong to one instance. None of them can see depth: two reconstructions that agree on every instance but disagree about which filament passes in front at every crossing are indistinguishable to all of them. The 2.5-D stage is therefore scored by its own family, defined here.

Crossing order accuracy. Each reference crossing names the instance physically in front; the reconstruction either names one or abstains. Three quantities are reported and none is quoted alone: *decided only*, the fraction of committed decisions that are correct, which by itself is not evidence because abstaining on every hard crossing scores 1.000; *all crossings*, which counts an abstention as an error and is the headline; and the *abstention rate* beside both. Chance is exactly 0.5, the decision being binary and symmetric, so only the margin above it carries information. A predicted crossing is credited only when it involves the same instance pair as a reference crossing and lies within the placement tolerance, so inventing crossings cannot help.

Depth order over instance pairs. Crossing accuracy is local. The global counterpart asks, for two instances, whether the reconstruction places the deeper one deeper, and is reported twice. Over *crossing pairs* it measures the stack where evidence exists. Over *all pairs* it includes instances that never cross and are therefore constrained by no evidence at all, whose relative depth is fixed by the solver's tie-break rather than by the image. The two are never pooled, and the fraction of pairs sharing a component is reported beside them so the second can be read against what it could possibly measure. Both are invariant to monotone rescaling of depth, so they describe the arrangement rather than its units. Kendall's τ over the same pairs is $2a - 1$ for accuracy a and is not reported separately.

Layer structure. Layers are an ordinal quotient of depth: a reconstruction can order every pair correctly yet cut the film into a different number of layers. Layer agreement, the fraction of instances placed in the reference layer, is therefore reported separately.

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